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Executive Summary

This report presents an evaluation of a dual-indicator trading strategy applied to a universe of nine Layer 1 cryptocurrencies. The strategy combines a Dual Hull Moving Average (HMA) crossover with a Z-score-based spread reversion signal filtered by slope flattening. The strategy also includes a trailing stop-loss and factors for trading cost. Back-testing was conducted across three timeframes **Daily, 5-minute, and 4-hour** to assess the robustness and effectiveness of the model under varying market regimes and to test models adaptability across timeframes. **The performance evaluation incorporated key metrics including Net Returns, Sharpe Ratios, Win Rates, Drawdowns, and Alpha (vs. Buy-and-Hold Benchmarks) through regression analysis. Daily timeframe delivered the most stable alpha profiles**, with multiple coins showing statistically significant positive alpha, indicating the strategy's ability to outperform passive holding. All detailed trade logs, equity curves, and regression visuals are included in the annex for review.

Data Collection

The study focuses on a curated basket of **nine major Layer 1 blockchain tokens: BTC, ETH, SOL, AVAX, XRP, ADA, BNB, SUI, and MATIC** (only for daily timeframe due to data availability constraints). The data was sourced from two platforms i.e. Yahoo Finance and Kraken via the `yfinance` & `ccxt` api library respectively. The period of study in the daily time frame is from 01/01/2020 to current date which was further split into modelling period of 01/01/2020 to 12/31/2024 and a test period from 01/01/2025 to current. For intraday analysis the data is taken for last 3 months in both 4hrs and 5 min time frame.

Data Cleaning & Validation Process

The raw data was processed into close only time series data for signal generation and following cleaning steps were applied :

Missing Data Handling

The coins with missing data or NAN's was cleaned using `dropna()` command. Due to the different originations/ launch of the coins we also needed to take into account date and time stamps in order to generate simultaneous signals.

Strategy Design

The model combines trend following and detection logic with mean reversion logic to develop trade signals and avoid false signals.

Core Logic

The strategy uses the Dual HMA framework to analyse and identify short term trend deviations. The signals are generated at the crossover point using a fast moving HMA and a slow moving HMA. As a crossover is detected the spread analysis is conducted. The spread analysis (fast hma – slow hma) uses the mean and standard deviation of last 25 days with the view of factoring any large news based spikes in the very recent trading month or the active window of 25 days. With this a trailing stoploss of 2.5% is also set up.

Signal Generation Process

The primary momentum filter uses two HMA's, selected for their reduced lag and smoothness. A fast HMA (period = 5) captures short-term trend shifts, while a slow HMA (period = 75) defines the directional view. Mean-reversion is detected via a Z-score of the spread exceeding ± 1 standard deviation indicates significant deviation serving as long or short entry signals contingent on trend confirmation. To manage exits a slope-flattening filter is used which monitors the absolute change in the fast HMA. If slope momentum decays below a 0.0005 threshold, the position is exited on trend exhaustion grounds. A 2.5% trailing stop-loss acts as a secondary exit mechanism, ensuring downside containment.

Transaction costs are fully incorporated, with a 0.40% round-trip assumption (0.20% per leg), applied prior to net return calculations.

Trade Execution Rules

Trades are initiated only when both momentum and mean-reversion filters align. A long position is entered when the Z-score falls below -1 and the fast HMA is above the slow HMA, indicating upward trend alignment. Shorts are initiated symmetrically when the Z-score exceeds +1 and the trend bias is negative.

Exit logic is either through slope flattening (loss of momentum) or trailing stop-loss breach. Positions are sized equally across assets with no leverage. Re-entry is disabled until a full exit has occurred and a fresh signal is generated.

Back-Testing Framework

The strategy is tested across three timeframes i.e. daily, 4hrs, 5min. Each time frame was tested independently but by using the same parameters. Following assumption was considered while testing: 40bps of trading costs for a full round trade signifying 20bps per leg.

Following output matrices was computed including: win rate, net and gross returns, Sharpe ratios, standard deviations, max drawdown and alpha. Apart from this a full trade log was generated to calculate time in trade and entry and exit levels and reasons.

Performance Results

Summary Returns in Daily Time Frame

Coin	Avg Net Return in %	Number of Trades
BTC – USD	0.1124	175
ETH – USD	0.1613	198
SOL – USD	2.3751	228
XRP – USD	1.1060	227
AVA X– USD	2.3504	201
MATIC – USD	0.4602	252
ADA – USD	1.3655	230
SUI – USD	-2.9359	142

Summary Metrics Across Timeframes

Timeframe	Avg. Net Return (%)	Avg. Sharpe Ratio	Avg. Win Rate (%)	Max Drawdown (%)	Remarks
Daily	Positive (most)	> 1.0 (select coins)	50–65%	Controlled	Most robust performance overall
5-Minute	Negative (majority)	< 0.5	~45%	High	Frequent exits + trading cost drag
4-Hour	Mixed	0.3 – 0.8	~50%	Moderate	Intermediate effectiveness

All returns are net of a 0.40% round-trip trading cost and incorporate a 2.5% trailing stop-loss.

Daily Alpha Analysis (Modelling Period)

Asset	Alpha	Beta	t-Stat (α)	p-Value (α)	R ²
ADA-USD	0.00219	0.13	2.25	0.025	6.10%
AVAX-USD	0.00443	0.16	3.15	0.0017	10.2%
SOL-USD	0.00526	0.03	3.78	0.00017	0.34%
XRP-USD	0.00251	0.07	2.16	0.031	2.49%
SUI-USD	-0.03065	0.21	-2.63	0.009	14.1%

During the modelling period (2020–2024), the strategy generated statistically significant alpha in ADA, AVAX, SOL, and XRP at the 5% level. AVAX and SOL stood out with 1% significance indicating outperformance.

In contrast, SUI-USD produced negative yet significant alpha, suggesting model incompatibility with its market behaviour.

Daily Alpha Analysis (Test Period)

Asset	Alpha	Beta	t-Stat (α)	p-Value (α)	R ²
ADA-USD	0.00098	0.24	0.91	0.362	8.24%
AVAX-USD	0.00010	0.36	0.04	0.965	13.0%
XRP-USD	0.00113	0.28	0.93	0.356	5.77%

In the 2025 test window, alpha decayed but remained directionally positive for several coins. However, no results reached statistical significance, suggesting either regime shift effects or reduced edge under recent volatility structures.

Intraday Performance: 5-Minute Interval

The 5-minute timeframe exhibited structural underperformance. While trade frequency increased, many positions were exited prematurely due to slope flattening, limiting trend capture. Combined with frictional trading costs, this led to predominantly negative net returns and no statistically significant alpha across assets.

High-frequency signal execution may require adaptive exit rules or microstructure-aware filters to restore viability

Performance 4-Hour

Asset	Alpha	Beta	t-Stat (α)	p-Value (α)	R ²
BNB-USD	0.566	64.23	1.10	0.282	2.76%
ADA-USD	0.518	-65.10	0.51	0.610	2.01%
SUI-USD	1.162	-64.08	0.89	0.378	1.55%
BTC-USD	-1.025	-25.45	-1.64	0.109	0.33%

The 4-hour window gave mixed results. Alpha magnitudes were higher, and some coins (e.g., BNB, ADA) showed directional outperformance, though without statistical significance. BTC-USD produced marginally negative alpha. Overall, the 4-hour data sits between daily and 5-minute profiles in terms of signal integrity and noise resilience.

Statistically Significant Alpha at 1% Confidence Level

The strategy delivered high-confidence outperformance in AVAX-USD and SOL-USD, both of which exhibited statistically significant positive alpha at the 1% level, with p-values of 0.0017 and 0.00017, respectively. AVAX demonstrated stable excess returns with moderate beta exposure, while SOL's alpha emerged alongside minimal beta (0.03), implying low correlation to the benchmark and potential diversification benefits.

Statistically Significant Alpha at 5% Confidence Level

At the 5% significance ADA-USD and XRP-USD also produced positive and statistically reliable alpha, with p-values of 0.025 and 0.031, respectively. Although the effect sizes were smaller, the directional consistency across both tokens reinforces the success of the combined HMA-Z -score framework. T

Intraday Performance and Limitations

5-Minute Timeframe Analysis

While the high-frequency data provided ample trade opportunities across most coins, the vast majority of trades were exited abruptly due to the **slope flattening condition**. This led to the majority of trades capturing only small price

movements, which were frequently **overwhelmed by the 0.40% round-trip transaction cost**. Other key observations were high number of trades were getting generated in sideways market due to which the stop filter reacted negatively and no alpha was generated in fact the trades also resulted into losses.

Hour Timeframe Analysis

Compared to the 5-minute data, the 4-hour timeframe offered **a more balanced trade-off between signal stability and responsiveness**. The number of trades was lower, but trade quality improved in terms of reduced drawdowns and slightly improved Sharpe Ratios. Some assets showed slight positive alpha but none which was statically significant.

Optimization and Future Improvements

The goal is to improve quality of signals generated along with increase of alpha and ensuring we do not attract more costs and slippage while execution.

Redefining Exit Logic

One of the key aspect identified causing trades to close before a full spectrum of effects was seen is the exit logic. We would like to see a reaction to a moderately sensitive slope feature which is used to determine exit.

Dynamic Thresholds for Z Score

The current ± 1 Z-score threshold is fixed across all coins and timeframes. However, asset-specific volatility and coin behaviour suggest that a **rolling or adaptive threshold** (e.g., based on realized volatility or signal dispersion) could increase signal precision.

Time Frame Specific Recommendation

Rather than using blanket parameters for all the time frames we recommend use of different configurations for the fast and slow calculation which could be per coin specific or time frame specific or a mix of both. Also we recommend that the z score window should be adjusted as per time frame as it can be used to factor geographic market open and close impact on volumes and prices.

Conclusion

In the light of above analysis the strategy shows clear signals and profitability in longer durations i.e. mainly for swing trades using daily time frame earning a steady average return across number of trades. With due considerations to the

future improvements the strategy can be furthered into an intraday high frequency scalping strategy.

References

Attribution for packages: ccxt, pandas, statsmodels, etc.

HMA documentation by Fidelity

<https://www.fidelity.com/learning-center/trading-investing/technical-analysis/technical-indicator-guide/hull-moving-average>

Annex

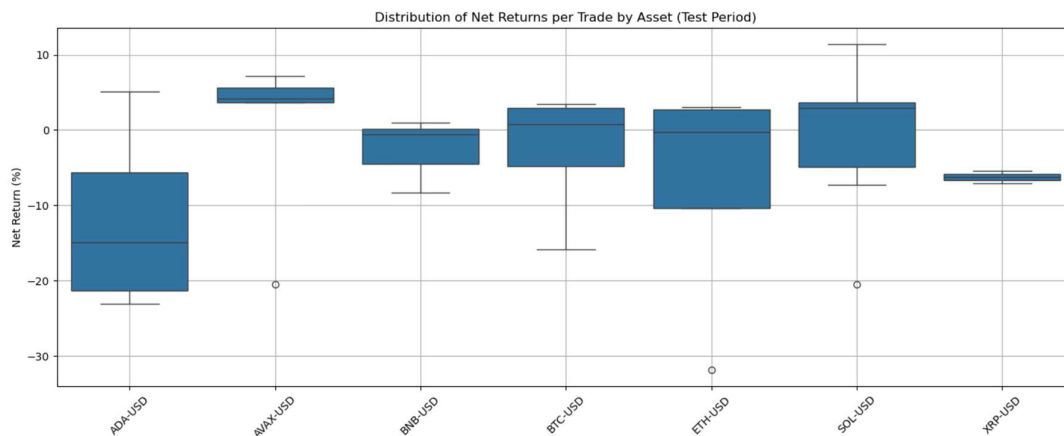
Trade log for daily time frame

	Trade Type	Entry Date	Entry Price	Exit Date	Exit Price	Gross Return (%)	Net Return (%)	Duration (days)	Exit Reason	Asset
0	Long	2020-04-15	0.03	2020-04-16	0.03	7.99	7.59	1	Slope Flattening	ADA-USD
1	Long	2020-04-17	0.03	2020-04-20	0.03	-1.15	-1.55	3	Trailing SL	ADA-USD
2	Long	2020-04-23	0.04	2020-04-30	0.05	17.83	17.43	7	Trailing SL	ADA-USD
3	Long	2020-05-01	0.05	2020-05-02	0.05	-0.48	-0.88	1	Slope Flattening	ADA-USD
4	Short	2020-05-25	0.05	2020-05-26	0.05	1.09	0.69	1	Slope Flattening	ADA-USD
...	...	...	...	...	...	...	...	...	...	...
1845	Short	2025-06-03	2.25	2025-06-06	2.16	3.90	3.50	3	Trailing SL	XRP-USD
1846	Short	2025-06-07	2.18	2025-06-08	2.27	-3.84	-4.24	1	Trailing SL	XRP-USD
1847	Short	2025-06-13	2.15	2025-06-16	2.24	-4.14	-4.54	3	Trailing SL	XRP-USD
1848	Short	2025-06-22	2.01	2025-06-23	2.17	-7.03	-7.43	1	Trailing SL	XRP-USD
1849	Long	2025-06-30	2.24	2025-07-01	2.17	-2.97	-3.37	1	Trailing SL	XRP-USD

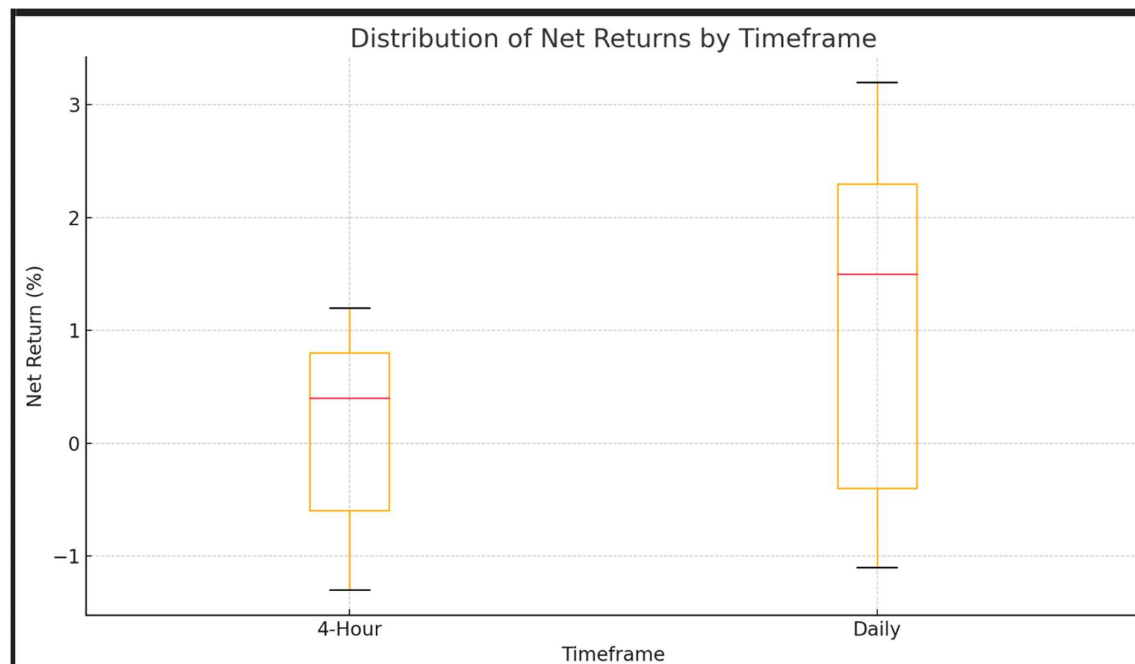
1850 rows x 10 columns

4	Short	2025-04-23	22.3385	2025-04-30	20.9094	6.83	6.43	7	Trailing SL	AVAX-USD
5	Long	2025-05-03	20.5481	2025-05-08	22.1101	7.60	7.20	5	Trailing SL	AVAX-USD
6	Short	2025-05-10	25.8884	2025-05-11	24.7523	4.59	4.19	1	Trailing SL	AVAX-USD
7	Short	2025-05-12	24.8320	2025-05-15	23.4339	5.97	5.57	3	Trailing SL	AVAX-USD
8	Long	2025-05-17	22.1936	2025-05-18	23.0896	4.04	3.64	1	Trailing SL	AVAX-USD
9	Long	2025-05-19	22.2728	2025-05-21	23.1880	4.11	3.71	2	Trailing SL	AVAX-USD
10	Long	2025-05-25	23.4417	2025-06-29	18.7416	-20.05	-20.45	35	Crossover	AVAX-USD
11	Short	2025-06-30	17.9749	2025-07-01	17.1056	5.08	4.68	1	Trailing SL	AVAX-USD
12	Short	2025-07-03	18.5391	2025-07-04	17.7951	4.18	3.78	1	Trailing SL	AVAX-USD
13	Short	2025-04-23	605.8150	2025-05-05	597.6898	1.36	0.96	12	Crossover	BNB-USD
14	Short	2025-05-07	603.2797	2025-05-30	655.4822	-7.96	-8.36	23	Crossover	BNB-USD
15	Long	2025-05-31	658.5013	2025-06-30	657.0253	-0.22	-0.62	30	Crossover	BNB-USD
16	Short	2025-04-21	87518.9062	2025-05-16	103489.2891	-15.43	-15.83	25	Crossover	BTC-USD
17	Long	2025-05-17	103191.0859	2025-05-18	106446.0078	3.15	2.75	1	Trailing SL	BTC-USD
18	Long	2025-05-19	105606.1797	2025-05-21	109678.0781	3.86	3.46	2	Trailing SL	BTC-USD
19	Long	2025-05-24	107791.1562	2025-06-26	106960.0000	-0.77	-1.17	33	Crossover	BTC-USD
20	Short	2025-04-22	1757.3317	2025-05-26	2564.1384	-31.47	-31.87	34	Crossover	ETH-USD
21	Long	2025-05-30	2529.9436	2025-06-02	2607.0974	3.05	2.65	3	Trailing SL	ETH-USD
22	Long	2025-06-03	2593.2803	2025-06-09	2681.5171	3.40	3.00	6	Trailing SL	ETH-USD
23	Long	2025-06-13	2579.4861	2025-07-04	2508.5183	-2.75	-3.15	21	Crossover	ETH-USD
24	Short	2025-04-18	134.0735	2025-05-04	143.9831	-6.88	-7.28	16	Crossover	SOL-USD

Returns distribution for top coins during test period



Return distribution across coins



Trade log 5 min

	Entry Date	Entry Price	Exit Date	Exit Price \
0	2025-07-12 19:25:00	117515.9000	2025-07-12 19:30:00	117499.9000
1	2025-07-12 19:35:00	117548.7000	2025-07-12 19:40:00	117589.7000
2	2025-07-12 19:55:00	117570.2000	2025-07-12 20:00:00	117564.7000
3	2025-07-12 20:05:00	117534.0000	2025-07-12 20:10:00	117482.3000
4	2025-07-12 20:15:00	117464.9000	2025-07-12 20:20:00	117482.1000
..	...	...	...	...
295	2025-07-14 20:25:00	3.8453	2025-07-14 20:30:00	3.8485
296	2025-07-14 20:35:00	3.8450	2025-07-14 20:40:00	3.8553
297	2025-07-14 20:45:00	3.8542	2025-07-14 20:50:00	3.8594
298	2025-07-14 20:55:00	3.8555	2025-07-14 21:00:00	3.8474
299	2025-07-14 21:20:00	3.8546	2025-07-14 21:25:00	3.8486

	Exit Reason	Trade Type	Gross Return (%)	Net Return (%) \
0	Slope Flattening	Long	-0.013615	-0.413615
1	Slope Flattening	Long	0.034879	-0.365121
2	Slope Flattening	Long	-0.004678	-0.404678
3	Slope Flattening	Long	-0.043987	-0.443987
4	Slope Flattening	Long	0.014643	-0.385357
..	...	...	...	...
295	Slope Flattening	Short	-0.083149	-0.483149
296	Slope Flattening	Short	-0.267165	-0.667165
297	Slope Flattening	Short	-0.134736	-0.534736
298	Slope Flattening	Short	0.210532	-0.189468
299	Slope Flattening	Short	0.155901	-0.244099

	Duration (minutes)	Asset
0	5.0	BTC-USD
1	5.0	BTC-USD
2	5.0	BTC-USD
3	5.0	BTC-USD
4	5.0	BTC-USD
..	...	...
295	5.0	SUI-USD
296	5.0	SUI-USD
297	5.0	SUI-USD

Trade log 4hrs coin – ETH – USD

2025-07-09 00:00:00	Crossover	ETH-USD		Trade Type	Entry Price
Entry Date					
2025-04-04 16:00:00	Short	122.61	2025-04-06 16:00:00		108.07
2025-04-06 20:00:00	Long	105.85	2025-04-08 00:00:00		111.28
2025-04-09 16:00:00	Short	118.92	2025-04-10 00:00:00		115.44
2025-04-10 04:00:00	Short	114.74	2025-04-10 12:00:00		109.57
2025-04-11 04:00:00	Short	116.42	2025-04-14 20:00:00		129.29
2025-04-15 00:00:00	Long	130.58	2025-04-19 00:00:00		138.81
2025-04-20 04:00:00	Short	139.61	2025-04-20 12:00:00		135.97
2025-04-20 16:00:00	Long	137.18	2025-04-21 08:00:00		139.12
2025-04-21 20:00:00	Long	136.63	2025-04-22 08:00:00		140.45
2025-04-22 12:00:00	Short	145.46	2025-04-26 00:00:00		151.41
2025-04-26 04:00:00	Long	151.42	2025-05-01 00:00:00		148.72
2025-05-01 04:00:00	Short	149.95	2025-05-05 00:00:00		144.93
2025-05-05 20:00:00	Short	146.61	2025-05-06 12:00:00		143.07
2025-05-06 16:00:00	Long	144.86	2025-05-06 20:00:00		146.82
2025-05-07 00:00:00	Short	146.23	2025-05-12 00:00:00		174.75
2025-05-12 04:00:00	Long	176.05	2025-05-13 16:00:00		182.31
2025-05-14 08:00:00	Long	181.07	2025-05-18 12:00:00		176.21
2025-05-18 16:00:00	Short	166.76	2025-05-19 08:00:00		162.26
2025-05-19 20:00:00	Short	166.85	2025-05-25 04:00:00		172.32
2025-05-25 08:00:00	Long	172.00	2025-06-01 20:00:00		157.74
2025-06-02 00:00:00	Short	155.62	2025-06-05 16:00:00		146.18
2025-06-06 00:00:00	Long	146.96	2025-06-07 12:00:00		151.05
2025-06-08 20:00:00	Short	152.53	2025-06-12 04:00:00		158.85
2025-06-12 08:00:00	Long	157.72	2025-06-16 00:00:00		155.59
2025-06-16 04:00:00	Short	157.34	2025-06-16 20:00:00		150.67
2025-06-20 00:00:00	Short	145.53	2025-06-20 20:00:00		140.15
2025-06-21 00:00:00	Long	140.81	2025-06-23 20:00:00		144.62
2025-06-24 00:00:00	Short	143.34	2025-06-30 08:00:00		150.46
2025-06-30 20:00:00	Short	154.85	2025-07-01 04:00:00		150.23
2025-07-01 08:00:00	Long	148.80	2025-07-06 20:00:00		151.91
2025-07-07 00:00:00	Short	151.92	2025-07-12 16:00:00		160.70
2025-07-12 20:00:00	Long	160.73	2025-07-14 00:00:00		167.10

Alpha analysis daily

	Asset	Alpha	Beta	t-Stat Alpha	p-Value Alpha	R-squared \
0	ADA-USD	0.002189	0.131594	2.245707	0.025013	0.060997
1	AVAX-USD	0.004433	0.160397	3.152378	0.001699	0.101997
2	BNB-USD	0.000004	0.188548	0.005582	0.995547	0.149303
3	BTC-USD	-0.000079	0.095414	-0.159759	0.873107	0.034594
4	ETH-USD	-0.000151	0.086103	-0.204342	0.838146	0.029332
5	MATIC-USD	0.000733	0.047632	0.497702	0.618863	0.008804
6	SOL-USD	0.005257	0.030597	3.776480	0.000173	0.003364
7	SUI-USD	-0.030646	0.213903	-2.629239	0.009455	0.141114
8	XRP-USD	0.002512	0.068201	2.162952	0.030888	0.024919

	Num Obs
0	752
1	609
2	784
3	880
4	707
5	651
6	675
7	151
8	692

Alpha analysis 4hrs

	Asset	Alpha	Beta	R-squared	t-stat (alpha)	p-value (alpha)	\
0	BTC-USD	-1.025183	-25.447070	0.003295	-1.641653	0.108502	
1	AVAX-USD	-0.008276	-13.276422	0.002124	-0.008081	0.993580	
2	BNB-USD	0.566014	64.227376	0.027561	1.097032	0.281973	
3	XRP-USD	-0.583327	3.525874	0.000055	-0.673444	0.503537	
4	ADA-USD	0.518222	-65.097159	0.020067	0.512780	0.610122	
5	SOL-USD	0.188587	20.436379	0.009142	0.267405	0.790045	
6	SUI-USD	1.161597	-64.082117	0.015493	0.888350	0.378022	
7	ETH-USD	0.031328	-80.055223	0.030237	0.031620	0.974892	

	N Trades
0	42
1	60
2	30
3	56
4	58
5	64
6	60
7	56